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CO1 – Assessment Test 3 (AT3)

Visualization Design Exercise

Question 1: Retail Sales Visualization

code

```
retail <- data.frame(  
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",  
    "B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),  
  Region=c("North","South","East","West","North","South","East","West",  
    "North","South","East","West","North","South","East","West",  
    "North","South","East","West"),  
  Category=c("Electronics","Furniture","Clothing","Food","Electronics",  
    "Furniture","Clothing","Food","Electronics","Furniture",  
    "Clothing","Food","Electronics","Furniture","Clothing",  
    "Food","Electronics","Furniture","Clothing","Food"),  
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,  
    560000,670000,195000,980000,430000,760000,280000,1150000,  
    510000,690000,220000,1250000),  
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,  
    72000,98000,-8000,205000,51000,132000,24000,250000,  
    69000,115000,15000,285000),  
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,  
    2100,3300,1100,4700,1750,3600,1450,5600,  
    2200,3900,1600,6100)  
)  
  
colors <- c("red","blue","green","orange")  
col <- colors[as.numeric(as.factor(retail$Region))]  
pch <- c(16,17,15,18)[as.numeric(as.factor(retail$Category))]  
sizes <- retail$Customers/1000  
  
plot(log10(retail$Sales), retail$Profit,  
  col=col,
```

```

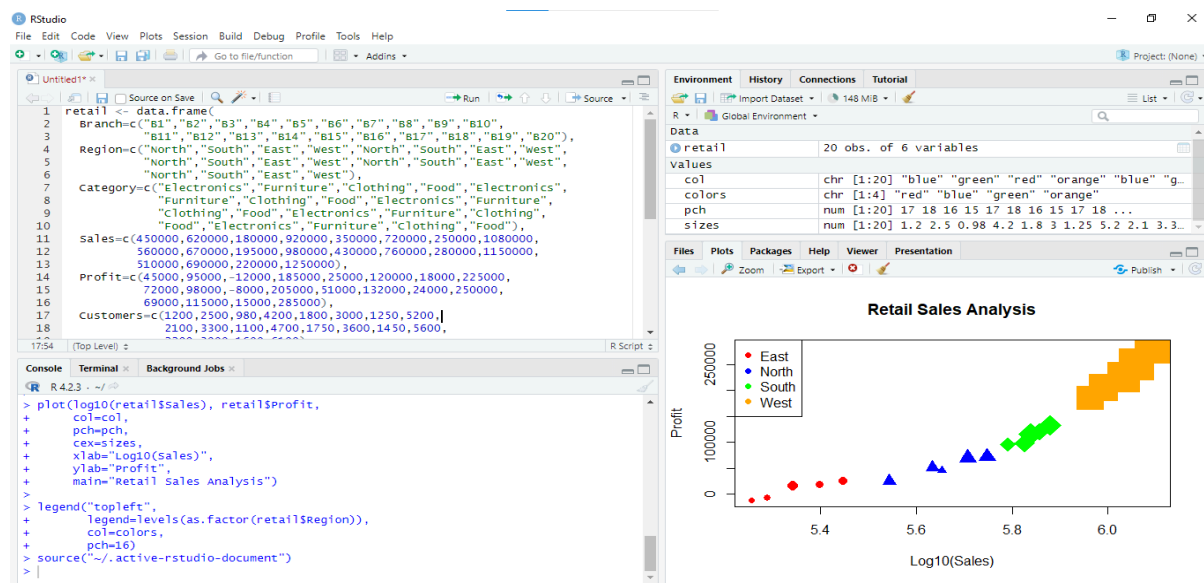
pch=pch,
cex=sizes,
xlab="Log10(Sales)",
ylab="Profit",
main="Retail Sales Analysis")

```

```

legend("topleft",
legend=levels(as.factor(etail$Region)),
col=colors,
pch=16)

```



Explanation

- A scatter plot is used to visualize the relationship between Sales and Profit.
- The log10 transformation is applied to Sales to reduce skewness and improve readability.
- Different colors represent different Regions.
- Different symbols (pch) represent Product Categories.
- Bubble size (cex) represents the number of Customers.
- The legend helps identify the region represented by each color.

Question 2: Electricity Generation Visualization

R Code

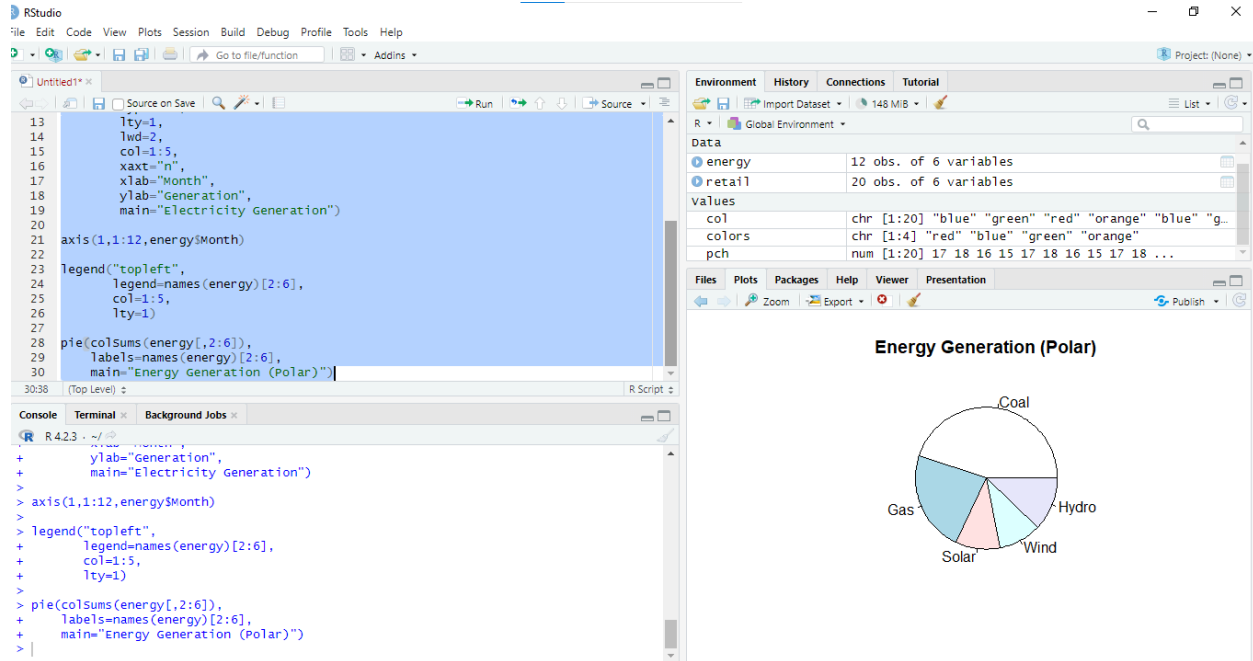
```
energy <- data.frame(  
  Month=month.abb,  
  Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),  
  Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),  
  Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),  
  Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),  
  Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)  
)
```

```
matplot(1:12,  
  energy[,2:6],  
  type="l",  
  lty=1,  
  lwd=2,  
  col=1:5,  
  xaxt="n",  
  xlab="Month",  
  ylab="Generation",  
  main="Monthly Electricity Generation")
```

```
axis(1,1:12,energy$Month)
```

```
legend("topleft",  
  legend=names(energy)[2:6],  
  col=1:5,  
  lty=1)
```

```
pie(colSums(energy[,2:6]),  
  labels=names(energy)[2:6],  
  main="Electricity Generation by Source")
```



Explanation

- Line chart (Cartesian coordinate) displays monthly trends for each energy source.
- Pie chart (Polar coordinate) represents the contribution of each energy source.
- The Cartesian graph is better for analyzing trends over time.
- The Polar graph is useful for showing the proportion of total energy generation.

Question 3: Climate Data Visualization

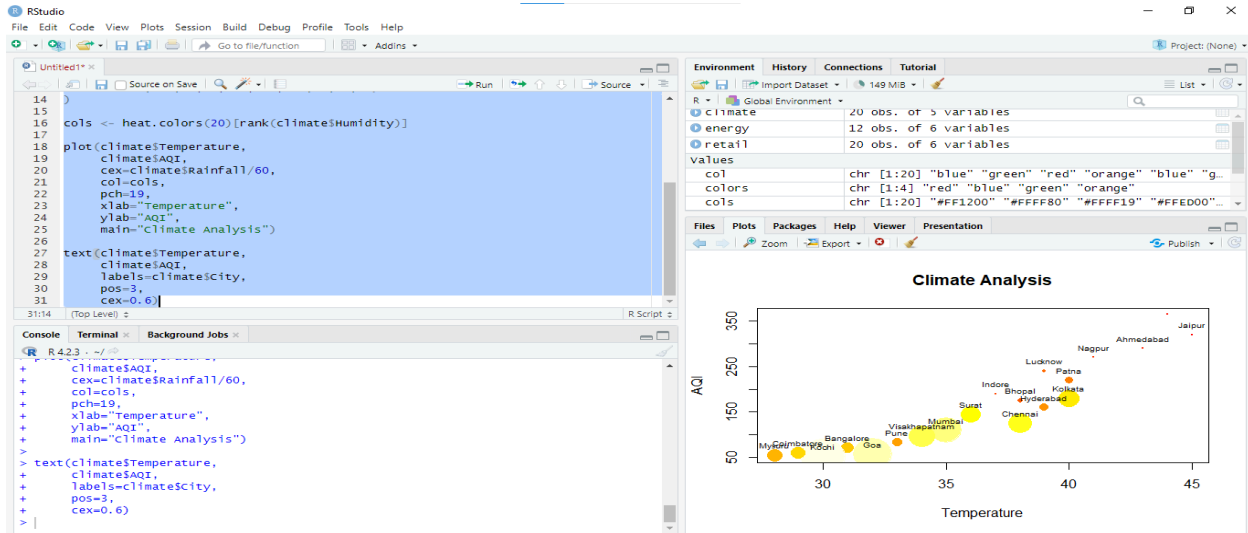
R Code

```
climate <- data.frame(  
  City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",  
    "Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",  
    "Bhopal","Patna","Goa","Kochi","Nagpur","Indore",  
    "Surat","Visakhapatnam","Coimbatore","Mysuru"),  
  Temperature=c(44,35,38,40,39,31,33,43,45,39,  
    38,40,32,30,41,37,36,34,29,28),  
  Humidity=c(22,84,78,72,48,66,58,26,18,42,  
    36,45,89,91,30,34,76,82,68,64),  
  Rainfall=c(6,280,220,190,85,120,95,2,1,35,  
    48,72,360,410,12,20,180,240,130,145),  
  AQI=c(365,110,125,180,160,72,84,290,320,240,  
    175,220,58,52,270,190,145,98,60,55)  
)
```

```
cols <- heat.colors(20)[rank(climate$Humidity)]
```

```
plot(climate$Temperature,  
  climate$AQI,  
  cex=climate$Rainfall/60,  
  col=cols,  
  pch=19,  
  xlab="Temperature",  
  ylab="AQI",  
  main="Climate Analysis")
```

```
text(climate$Temperature,  
  climate$AQI,  
  labels=climate$City,  
  pos=3,  
  cex=0.6)
```



Explanation

- Scatter plot shows the relationship between Temperature and AQI.
- Color represents Humidity.
- Bubble size represents Rainfall.
- City names are displayed using the `text()` function.
- The heat color palette provides an easy way to distinguish humidity levels.

Question 4: Software Project Statistics

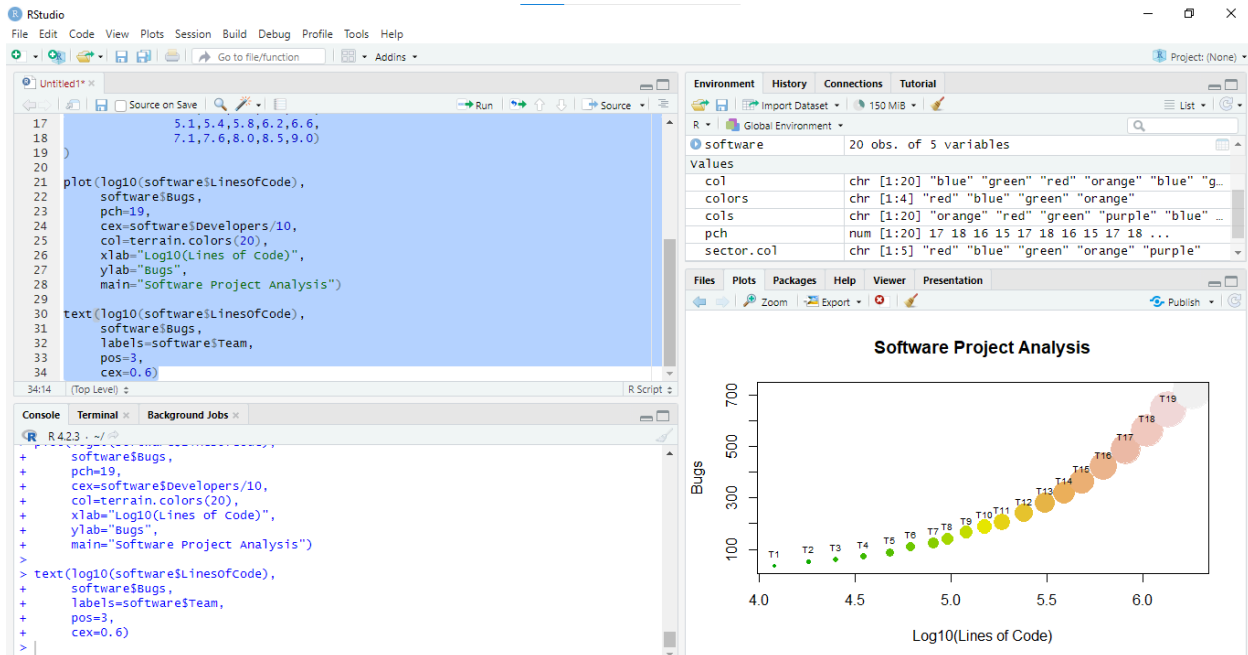
R Code

```
software <- data.frame(  
  Team=paste("T",1:20,sep=""),  
  LinesOfCode=c(12000,18000,25000,35000,48000,  
    62000,81000,96000,120000,150000,  
    185000,240000,310000,390000,480000,  
    620000,810000,1050000,1350000,1800000),  
  Bugs=c(35,52,60,72,88,  
    110,125,142,165,188,  
    205,240,280,320,365,  
    420,490,560,640,720),  
  Developers=c(5,7,9,10,12,  
    14,16,18,20,22,  
    25,28,30,34,38,  
    42,46,50,55,60),  
  Complexity=c(2.1,2.4,2.7,3.0,3.2,  
    3.5,3.8,4.1,4.5,4.8,  
    5.1,5.4,5.8,6.2,6.6,  
    7.1,7.6,8.0,8.5,9.0)  
)
```

```
plot(log10/software$LinesOfCode),  
software$Bugs,  
pch=19,  
cex=software$Developers/10,  
col=terrain.colors(20),  
xlab="Log10(Lines of Code)",  
ylab="Bugs",  
main="Software Project Analysis")
```

```
text(log10/software$LinesOfCode),  
software$Bugs,  
labels=software$Team,  
pos=3,
```

cex=0.6)



Explanation

- A scatter plot is used to analyze the relationship between Lines of Code and Bugs.
- A logarithmic transformation is applied to Lines of Code because the values are highly skewed.
- Bubble size represents the number of Developers.
- Different colors indicate variation among teams.
- Team names are displayed above each point.

Question 5: Company Performance Visualization

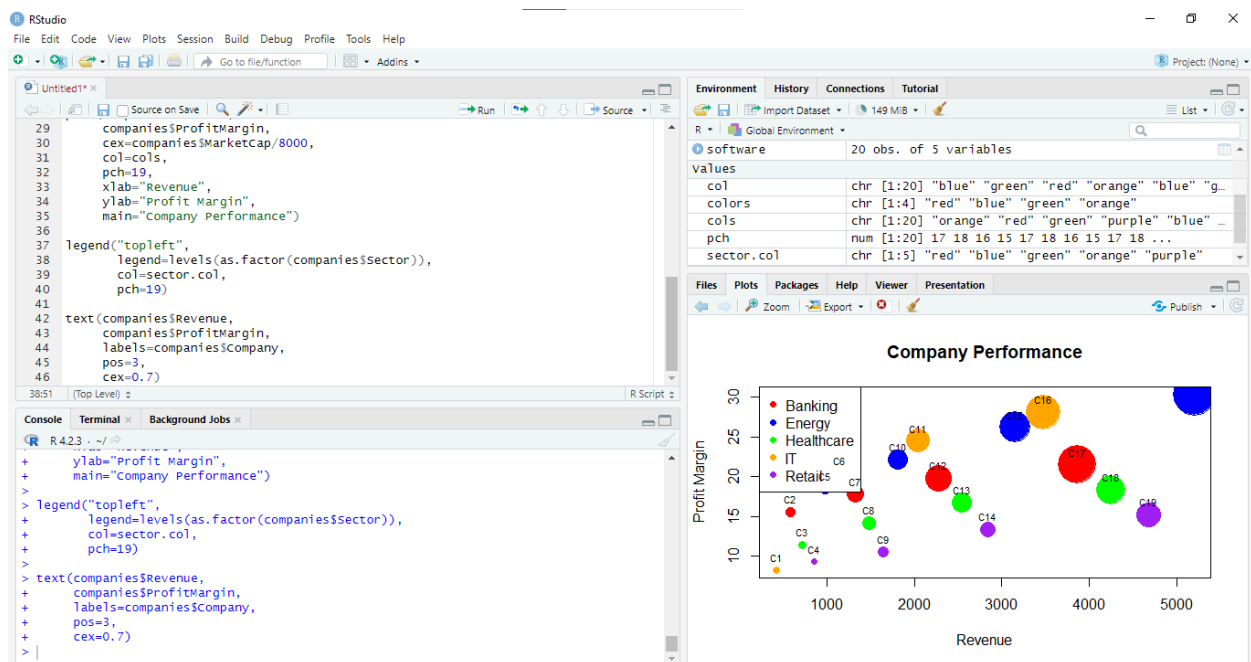
R Code

```
companies <- data.frame(  
  Company=paste("C",1:20,sep=""),  
  Sector=c("IT","Banking","Healthcare","Retail","Energy",  
            "IT","Banking","Healthcare","Retail","Energy",  
            "IT","Banking","Healthcare","Retail","Energy",  
            "IT","Banking","Healthcare","Retail","Energy"),  
  Revenue=c(420,580,720,850,980,  
            1150,1320,1480,1650,1820,  
            2050,2280,2550,2840,3150,  
            3480,3860,4250,4680,5200),  
  ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,  
                20.4,17.8,14.2,10.5,22.1,  
                24.5,19.8,16.7,13.4,26.2,  
                28.1,21.5,18.3,15.2,30.4),  
  Employees=c(1500,3200,2100,4200,2800,  
              5100,6800,3900,7200,4500,  
              8100,9500,6200,10300,7200,  
              11800,13200,8600,14500,16000),  
  MarketCap=c(8500,12000,9800,7600,14500,  
              18200,21500,16800,13200,24500,  
              28500,32500,24800,19200,38200,  
              42500,46800,35600,29800,52000)  
)  
  
sector.col <- c("red","blue","green","orange","purple")  
cols <- sector.col[as.numeric(as.factor(companies$Sector))]  
  
plot(companies$Revenue,  
     companies$ProfitMargin,  
     cex=companies$MarketCap/8000,
```

```
col=cols,
pch=19,
xlab="Revenue",
ylab="Profit Margin",
main="Company Performance")
```

```
legend("topleft",
legend=levels(as.factor(companies$Sector)),
col=sector.col,
pch=19)
```

```
text(companies$Revenue,
companies$ProfitMargin,
labels=companies$Company,
pos=3,
cex=0.7)
```



Explanation

- A scatter plot is used to compare Revenue and Profit Margin.
- Different colors distinguish Industry Sectors.
- Bubble size represents Market Capitalization.
- Company names are displayed beside each point.
- A legend is included to identify sector colors.

- This visualization effectively represents multiple variables while maintaining readability.